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CODE:YOU

PART 2: Log Analyzer Option

🧪 Capstone Part II – Track 1: Network Forensics (PCAP Analysis)

Assignment Title

Capstone Part II: Network Forensics Investigation (PCAP Analysis)

🏢 Scenario

You are a Security Operations Center (SOC) analyst investigating suspicious network activity within your organization.

A network-based alert has been triggered indicating potential malware-related activity originating from an internal host. The alert suggests possible communication with known malicious infrastructure.

You have been provided with a **packet capture (PCAP)** containing network traffic associated with this activity located here. ***The password to the .zip file containing the pcap is `infected`.***

Your task is to analyze this traffic and determine:

- whether a system is compromised
- how the compromise likely occurred
- what actions the attacker performed after gaining access

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report.

🌐 Environment Context

Assume the following about the environment:

- Internal network range: provided with the dataset
- Enterprise Windows-based environment
- Domain and host naming conventions may be visible in traffic
- One or more internal systems may be involved

> ⚠️ You will not have perfect or complete data. You must make reasonable, evidence-based conclusions.

🎯 Objective

Analyze the provided PCAP file and produce a **professional network forensic report**.

Your analysis must identify and explain:

- suspicious traffic patterns
- potential command-and-control (C2) activity
- DNS anomalies
- possible data exfiltration
- a reconstructed timeline of attacker behavior

🛠️ Tools

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- Wireshark (primary)
- tcpdump
- tshark (optional / advanced)

📦 Deliverables

You must submit:

1. Forensic Report (Required)

- `network_forensics_report.md` or `.pdf`

2. Timeline of Activity (Required)

- chronological sequence of events

3. Filters Used (Required)

- Wireshark display filters or command-line queries used during analysis

4. Indicators of Compromise (Required)

- IP addresses
- domains
- URLs
- file artifacts (if applicable)

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Answers without justification will not receive full credit.

📦 Report Structure (Required)

🔷 1. Executive Summary

Provide a high-level summary of the incident:

- What happened
- Severity of the issue
- Potential business impact

👉 Audience: non-technical stakeholders

🔷 2. Environment Overview

Describe what you can infer about the environment:

- internal network structure
- domain (if visible)
- types of systems present

🔷 3. Host Identification

Identify the system most likely involved in suspicious activity.

Include:

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- hostname (if available)
- associated user (if identifiable)

You must justify your identification using evidence.

🔷 4. Traffic Analysis Findings

A. Suspicious Traffic Patterns

- unusual communication behavior
- repeated connections
- beaconing patterns

B. DNS Analysis

- suspicious or uncommon domains
- patterns indicating malicious infrastructure

C. Command-and-Control (C2) Activity

- outbound communication to suspicious systems
- periodic or structured communication behavior

D. Data Exfiltration Analysis

- evidence of outbound data transfer
- abnormal request sizes or patterns

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List all identified indicators:

- IP addresses
- domains
- URLs
- file artifacts

Explain why each indicator is considered suspicious.

🔥 6. Timeline of Activity (Critical Section)

Reconstruct a timeline that includes:

1. initial suspicious activity
2. evidence of compromise
3. outbound communication
4. follow-on behavior

Your timeline must be:

- chronological
- supported by evidence

🔷 7. Attack Narrative

Explain:

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- initial infection
- execution
- command-and-control communication
- attacker activity

◆ 8. Impact Assessment

Explain:

- which systems are affected
- what data may be at risk
- potential organizational impact

◆ 9. Recommendations

Immediate Actions

- isolate affected systems
- block malicious IPs/domains
- reset credentials

Short-Term Improvements

- endpoint scanning
- patching

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- deploy EDR
- improve DNS monitoring
- implement stronger logging and alerting

🔷 10. Reflection

Briefly explain:

- what was challenging
- what techniques worked best
- what you would improve

📏 Requirements

- Length: **1000–1500 words**
- Must include:
  - timeline
  - IOCs
  - filters used
- Must include reasoning and evidence—not just answers

🧠 Grading Rubric

1. Technical Analysis – 30 points

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2. Timeline Reconstruction – 20 points

- logical sequence of events
- supported by evidence

3. IOC Identification – 15 points

- completeness
- accuracy
- justification

4. Attack Narrative – 15 points

- coherent explanation
- ties multiple findings together

5. Communication & Clarity – 10 points

- well-structured
- professional tone
- readable

6. Completeness – 10 points

- all required sections included

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Upload →

Upload a file to continue with this lesson.